

Binary Parry Sectors of the Tensor-Prime Shift: A Symbolic Gamma Factor and a Chiral No-Motion Theorem

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Abstract

The tensor-prime full-shift candidate reproduces the Euler product but did not previously generate its Archimedean factor. We isolate the binary full shift F_2 , the unique least nonunit tensor atom, and its Parry projection $K_2 = J_2/2$. The permutation representation $\mathbb{C}^2 = \mathbf{1} \oplus \text{sgn}$ gives two canonical channels. The stationary channel yields $\det(I - \mathcal{A}_s) = \prod_p (1 - p^{-s}) = \zeta(s)^{-1}$ for $\text{Re } s > 1$. The sign channel yields, along odd lengths,

$$\mathbb{E} \left| S_N / \sqrt{2\pi N} \right|^{s-1} \longrightarrow \pi^{-s/2} \Gamma(s/2)$$

locally uniformly for $\text{Re } s > 0$. Both are generated by the single tilted Parry trace identity $\text{tr}(e^{zQ/2} K_2 e^{zQ/2})^r = (\cosh z)^r$: $z = 0$ gives the Euler ledger, whereas $z = iu/\sqrt{r}$ gives the Gaussian characteristic function. This defines SD-C08 and a same-source Mellin–Fredholm factorization of $\pi^{-s/2} \Gamma(s/2) \zeta(s)$ on $\text{Re } s > 1$. It is not one dynamical determinant. We also prove that the one-sided Hellinger chiral ansatz is unitarily isospectral along $s = 1/2 + it$ for every bounded coupling, whether or not it commutes with tensor mass. Its natural $\det_3$ is s -independent and deletes the divergent quadratic trace. Exact computations confirm the identities and expose rank-one and central-limit universality. The stage reaches partial A3 but not global completion; Route B remains locked.

Research status. This exploratory note promotes SD-C08 at Route-A layer A3: the unique minimal binary atom generates the Euler and Archimedean channels through one tilted Parry trace family. It does not derive a unified dynamical determinant, meromorphic continuation, the functional equation, or a zero divisor. The stage therefore stops at global completion, and Route B remains locked.

1 Introduction

The arithmetic symbolic-dynamics program studied here begins from a rigid source rather than a target spectrum. Finite full shifts satisfy

$$F_m \boxtimes F_n \cong F_{mn}, \quad h_{\text{top}}(F_n) = \log n.$$

Tensor atoms therefore recover the rational primes as internal indecomposables, their entropy is the desired logarithmic clock, and loop repetition supplies prime powers. The resulting diagonal atom transfer gives the Euler product in its honest half-plane. Paper05 added an intrinsic exterior grading, but its strongest same-family completion attempts failed: they generated neither a half-density nor an Archimedean factor.

The present note takes a deliberately narrow next step. The tensor source has a unique least nonunit atom, F_2 . Its maximal-entropy Markov kernel is

$$K_2 = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

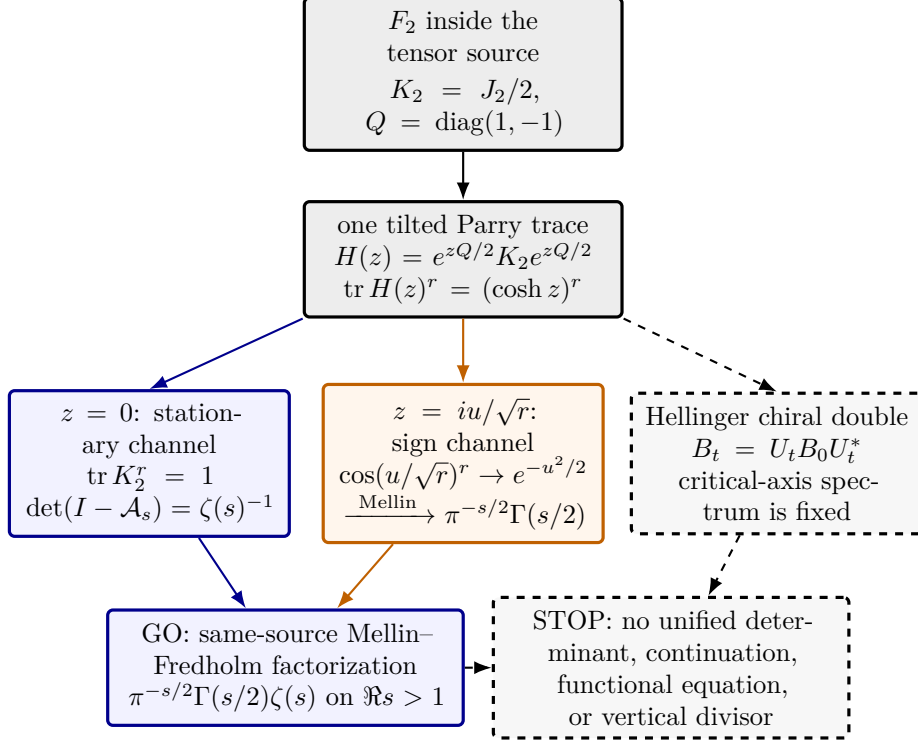


Figure 1: The two positive channels and the obstruction come from the same binary Parry source. Solid arrows are proved identities or limits; dashed arrows mark the chiral attempt and the global stop. Color is redundant with line style and labels.

More importantly, symbol exchange decomposes its permutation representation into two multiplicity-one lines. The trivial line is stationary, while the sign line is centered. These two sectors furnish an exact finite-prime channel and a fluctuation channel without adding a second system family.

Our key organizing identity uses the sign operator $Q = \text{diag}(1, -1)$ and the tilted Parry matrix

$$H(z) = e^{zQ/2} K_2 e^{zQ/2} = \frac{1}{2} \begin{pmatrix} e^z & 1 \\ 1 & e^{-z} \end{pmatrix}.$$

It has rank one and nonzero eigenvalue $\cosh z$, so

$$\text{tr } H(z)^r = (\cosh z)^r. \quad (1.1)$$

At $z = 0$, this is the unit power trace needed for every prime-power Euler coefficient. At $z = iu/\sqrt{r}$, it is the exact characteristic function of a normalized binary Birkhoff sum and tends to $e^{-u^2/2}$. The Mellin transform of the resulting self-dual Gaussian is $\pi^{-s/2} \Gamma(s/2)$. Thus the two channels are not coordinates copied from unrelated candidates: they are two specializations of one source-internal trace family.

This produces the new symbolic candidate SD-C08. In its initial common domain $\text{Re } s > 1$, its two canonical transforms give

$$\mathfrak{Z}_{\text{SD}}(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s).$$

The formula is a *Mellin–Fredholm factorization*. We do not call it a single dynamical determinant. We prove neither meromorphic continuation nor the functional equation; we do not insert $s(s-1)/2$; and we do not score zeros. This claim boundary is the difference between an A3 mechanism and a purported solution of RH.

The negative theorem is as useful as the positive construction. A natural Hellinger pairing replaces p^{-s} and $p^{-(1-s)}$ by a chiral block. On $s = 1/2 + it$ the paired weights have modulus $p^{-1/2}$, suggesting a centered self-adjoint family. We show that every one-sided implementation $A_t = G^{1/2+it}K$ with bounded K is unitarily conjugate to its value at $t = 0$, without any commutation hypothesis. The entire vertical parameter is a phase gauge. In particular, cross-atom noncommutation does not create spectral motion in this ansatz, even though it still risks manufacturing mixed primitive cycles and corrupting the exact Euler ledger. A live replacement must change the phase architecture itself while preserving that ledger.

The contributions are:

1. an intrinsic selection theorem for the binary scalar fluctuation sector inside the full-shift tensor source;
2. a single tilted-trace identity generating both the Euler and Gaussian channels;
3. locally uniform Mellin convergence to the Archimedean Gamma factor;
4. dimension-shifted q -symbol controls and explicit universality limitations;
5. a general critical-axis chiral no-motion theorem and an exact $\det_3$ loss formula;
6. a preregistered Route-A decision: A3 advances, global completion stops, and Route B remains locked.

All primary constructions remain in Symbolic Dynamics. Analogies with other system families are outside the scope of this note and are recorded, if at all, only as future clues.

2 Symbolic zeta, Parry measures, and fluctuation twists

The use of finite-state matrices to encode symbolic zeta functions goes back to the determinant formulas of [Bowen and Lanford \(1970\)](#); transfer-operator continuation in hyperbolic dynamics was developed in a much broader setting by [Ruelle \(1976\)](#). These results make a determinant representation natural, but they do not identify the tensor atoms used here or manufacture the Riemann Euler product from a full-shift monoidal source.

The maximal-entropy Markov measure for a finite-state shift is intrinsic to the transition structure ([Parry, 1964](#)). Normalized transfer operators and g -measures provide a standard language for this probabilistic sector ([Keane, 1972](#); [Walters, 1975](#)). We use only the elementary full two-shift case: its transition matrix is $K_2 = J_2/2$, and successive symbols are independent and uniform. Calling K_2 a Parry Markov fiber does not turn the matrix-weighted atom loop into a new skew-product dynamical system.

Central and local limit theorems for symbolic systems, as well as analytic twists of transfer operators, are classical ([Lalley, 1986](#); [Parry and Pollicott, 1990](#); [Coelho and Parry, 1990](#); [Aaronson and Denker, 2001](#)). In particular, one frequently studies a family obtained by multiplying a transfer operator by e^{zf} ; derivatives of its leading eigenvalue encode fluctuations. Our finite matrix $H(z)$ is the completely explicit two-state instance. The point of the present synthesis is not a new CLT. It is that the same tilted trace family has two arithmetic-symbolic specializations: its untwisted trace preserves the tensor-prime Euler ledger, and its diffusive imaginary twist generates the Gaussian whose Mellin transform is the Archimedean Gamma factor.

The literature audit supports the ingredients but not a broad priority claim. Within the finite primary-source search performed for this stage, we did not find the exact package consisting of the tensor-prime full-shift monoid, its minimal binary Parry sector, the same-trace Euler/Gamma specialization, and the universal one-sided chiral no-motion theorem. Absence from a finite search is not evidence of first-ever novelty, so our novelty claim is limited to this source-locked synthesis and its falsification boundary.

Finally, a fluctuation Gamma factor is generic unless its observable and dimension are selected. The binary sign representation is canonical up to orientation, but biased binary chains and scalar projections in larger alphabets still have Gaussian limits after centering and variance normalization. We retain those controls explicitly rather than presenting the Gamma limit as sufficient evidence on its own.

3 The minimal binary source and one tilted trace

Let $F_n = \{1, \dots, n\}^{\mathbb{Z}}$ with the shift map. We work with the symmetric monoidal skeleton generated by Cartesian product.

Proposition 3.1 (Intrinsic binary selection). *The unique nonunit tensor atom of minimum topological entropy is F_2 . Moreover, among the permutation representations of the full shifts, F_2 is the unique one whose centered standard representation is a canonical scalar line.*

Proof. The tensor-isomorphism classes form $(\mathbb{N}_{\geq 1}, \times)$ and entropy is $\log n$. Nonunit indecomposables are therefore F_p , and strict monotonicity of $\log n$ selects F_2 . For F_q , the symbol-permutation representation decomposes as

$$\mathbb{C}^q = \mathbb{C}\Omega_q \oplus E_q, \quad \Omega_q = q^{-1/2}(1, \dots, 1),$$

where $E_q = \Omega_q^\perp$ is the standard representation of dimension $q - 1$. It is one-dimensional exactly when $q = 2$. Its two unit generators differ only by sign, which disappears under an absolute moment. $\square$

The unique positive S_q -invariant unit vector is Ω_q , and its projection is

$$K_q = |\Omega_q\rangle\langle\Omega_q| = J_q/q.$$

For $q = 2$, introduce the sign involution $Q = \text{diag}(1, -1)$. Relabeling the two symbols sends Q to $-Q$ but leaves every even or absolute quantity below unchanged.

Theorem 3.2 (Tilted Parry trace). *For*

$$H(z) = e^{zQ/2} K_2 e^{zQ/2} = \frac{1}{2} \begin{pmatrix} e^z & 1 \\ 1 & e^{-z} \end{pmatrix},$$

one has, for every integer $r \geq 1$,

$$\text{tr } H(z)^r = (\cosh z)^r.$$

In particular, $z = 0$ gives $\text{tr } K_2^r = 1$, while

$$\text{tr } H(iu/\sqrt{r})^r = \cos(u/\sqrt{r})^r \longrightarrow e^{-u^2/2}.$$

Proof. The matrix $H(z)$ has rank one. Its trace, hence its only nonzero eigenvalue, is $(e^z + e^{-z})/2 = \cosh z$. The power-trace formula follows. The last limit is obtained from $r \log \cos(u/\sqrt{r}) = -u^2/2 + O(r^{-1})$ locally uniformly in u . $\square$

By cyclicity, these are also the power traces of the cyclic representative $H_{\text{cyc}}(z) = K_2 e^{zQ}$ (up to the harmless convention $Q \mapsto -Q$). On the active rank-one sector it has the same nonzero eigenvalue and Fredholm block determinant as the symmetric representative; the numerical implementation uses $H_{\text{cyc}}(iu)$.

The final expression is also exactly the characteristic function of $r^{-1/2} \sum_{j=1}^r X_j$ for independent $X_j \in \{-1, 1\}$. Thus a single finite symbolic trace family contains both channels before any Gamma function is mentioned.

3.1 Euler transfer as the untwisted channel

Let At denote the nonunit tensor atoms and set

$$\mathcal{H} = \ell^2(\text{At}) \otimes \mathbb{C}^2, \quad \mathcal{A}_{s,z} = \bigoplus_{F_p \in \text{At}} p^{-s} H(z).$$

This is a matrix-weighted transfer on the atom-loop symbolic suspension. It is not a group cocycle and we do not claim an independently constructed skew shift.

Theorem 3.3 (Exact stationary ledger). *For $\text{Re } s > 1$,*

$$\text{Tr } \mathcal{A}_{s,0}^r = \sum_p p^{-rs}, \quad \det(I - \mathcal{A}_{s,0}) = \prod_p (1 - p^{-s}) = \zeta(s)^{-1}.$$

With the fixed purely odd one-particle convention, $\text{Ber}(I - \mathcal{A}_{s,0}) = \zeta(s)$.

Proof. Since $K_2^r = K_2$ and $\text{tr } K_2 = 1$, each atom block contributes p^{-rs} . Trace-class convergence holds for $\text{Re } s > 1$, where the logarithmic Fredholm expansion is absolutely convergent:

$$\log \det(I - \mathcal{A}_{s,0}) = - \sum_{r \geq 1} \frac{1}{r} \sum_p p^{-rs} = \sum_p \log(1 - p^{-s}).$$

The Berezinian is the reciprocal determinant on a purely odd space. $\square$

The exact ledger selects rank one, not binary fairness. Uniform K_q and biased rank-one kernels have the same power traces. Conversely, for a finite reversible stochastic kernel, the identities $\text{tr } K^r = 1$ for all r force all nonstationary eigenvalues to vanish; reversibility then gives a rank-one orthogonal projection. Binary selection enters through the tensor minimum and the one-dimensional centered representation, not through the Euler identity alone.

4 From binary fluctuations to the Gamma factor

Let $(X_j)_{j \geq 0}$ be the sign observable along the Parry process of F_2 , so the X_j are independent and uniform on $\{-1, 1\}$. Put

$$S_N = \sum_{j=0}^{N-1} X_j, \quad Y_N = \frac{S_N}{\sqrt{2\pi N}}.$$

We take N odd. This removes the finite- N atom at zero, which would make negative moments undefined even though the limiting singularity is integrable.

Theorem 4.1 (Binary Archimedean limit). *For odd N and $\text{Re } s > 0$, define*

$$M_N(s) = \mathbb{E}|Y_N|^{s-1}.$$

Then, locally uniformly on the half-plane $\text{Re } s > 0$,

$$M_N(s) \longrightarrow \int_{\mathbb{R}} |x|^{s-1} e^{-\pi x^2} dx = \pi^{-s/2} \Gamma(s/2).$$

Proof sketch with the uniformity mechanism. The specialization in [theorem 3.2](#) gives the central limit theorem. A lattice local central limit estimate gives uniform Gaussian approximation on bounded windows, while the binomial tail estimate controls large $|Y_N|$. Near zero, the local estimate implies a uniform small-ball bound $\mathbb{P}(|Y_N| \leq x) \leq Cx + CN^{-1/2}$. For every compact $C \Subset \{\text{Re } s > 0\}$ this controls $|Y_N|^{\text{Re } s - 1} (1 + |\log |Y_N||^k)$ uniformly for the finitely

many s -derivatives needed by normal-family convergence. Truncation at small and large radii therefore upgrades weak convergence to locally uniform Mellin convergence. Finally, $u = \pi x^2$ gives

$$2 \int_0^\infty x^{s-1} e^{-\pi x^2} dx = \pi^{-s/2} \int_0^\infty u^{s/2-1} e^{-u} du.$$

□

The $1/2$ is now an internal fluctuation exponent, not a number copied from the target critical line. The constant π is the normalization for the self-dual Gaussian density $e^{-\pi x^2}$ under Fourier transform. Changing Gaussian convention would multiply the Mellin transform by a zero-free exponential in s , but we freeze the self-dual convention before comparison.

4.1 Representation-dimensional controls

For the uniform q -shift, place its q symbols at the vertices $v_1, \dots, v_q$ of a centered regular simplex in $E_q \cong \mathbb{R}^d$, $d = q - 1$, with $\|v_j\| = 1$ and $\langle v_j, v_k \rangle = -1/d$ for $j \neq k$. This uses the full standard representation and does not select a favorable scalar projection.

Proposition 4.2 (Radial q -symbol control). *If V_j are independent uniform simplex vertices and*

$$R_N = \frac{\|\sum_{j=1}^N V_j\|}{\sqrt{2\pi N}},$$

then, for $\operatorname{Re} s > 1 - d$ where the limiting moment is locally integrable,

$$\mathbb{E} R_N^{s-1} \longrightarrow M_d(s) := (\pi d)^{-(s-1)/2} \frac{\Gamma((d+s-1)/2)}{\Gamma(d/2)}.$$

For $d = 1$ this is $\pi^{-s/2} \Gamma(s/2)$; for every $d > 1$ the Gamma shift is different.

Proof. The simplex covariance is I_d/d , so $\sqrt{d/N} \sum_j V_j$ converges to a standard d -dimensional Gaussian Z . The radial moment formula

$$\mathbb{E} \|Z\|^{s-1} = 2^{(s-1)/2} \frac{\Gamma((d+s-1)/2)}{\Gamma(d/2)}$$

and the scale $R_N = \|\sqrt{d/N} \sum_j V_j\| / \sqrt{2\pi d}$ give the claim. □

This control restores a meaningful part of binary specificity: the canonical full standard representation has the exact Archimedean shape only in dimension one. It does not defeat every genericity challenge. A biased binary chain, after centering and variance normalization, has the same one-dimensional Gaussian limit; an arbitrarily selected scalar projection of F_q does too. Those choices are not the multiplicity-one S_q -representation supplied by the source, but their success means the Gamma limit alone cannot carry the arithmetic claim.

4.2 The same-source Mellin–Fredholm factorization

The stationary and sign channels are specializations of the same tilted trace family $H(z)$. Combining [theorems 3.3](#) and [4.1](#) therefore defines, on their common initial domain,

$$\mathfrak{Z}_{\text{SD}}(s) := \left(\lim_{N \rightarrow \infty} M_N(s) \right) \operatorname{Ber}(I - \mathcal{A}_{s,0}) = \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad \operatorname{Re} s > 1. \quad (4.1)$$

Equation (4.1) is an exact factorization on that half-plane. It is stronger than coordinate splicing because both transforms descend from the same multiplicity-free S_2 source and the same matrix trace deformation. It is weaker than a unified determinant because the Mellin limit and Berezinian are still distinct analytic operations. No continuation or functional equation is inferred from their product.

5 Hellinger centering and the chiral no-motion theorem

The factor $p^{-1/2}$ can also be generated without inserting a square root by hand. Pair the weights p^{-s} and $p^{-(1-s)}$ in an off-diagonal block; their product is p^{-1} . On the line $s = 1/2 + it$ the block is self-adjoint. This Hellinger-style pairing selects the center exactly, but exact centering does not imply spectral motion.

Let

$$G = \bigoplus_p p^{-1} I_{E_p}$$

on a Hilbert sum of finite atom fibers E_p . Let K be any bounded operator on that sum and define

$$A_t = G^{1/2+it} K, \quad B_t = \begin{pmatrix} 0 & A_t \\ A_t^* & 0 \end{pmatrix}.$$

Theorem 5.1 (Universal one-sided chiral no-motion). *For every real t ,*

$$B_t = U_t B_0 U_t^*, \quad U_t = \text{diag}(G^{it}, I).$$

Consequently the spectrum, singular values, and every unitarily invariant regularized determinant of B_t are independent of t . No commutation condition on G and K is required.

Proof. Put $A_0 = G^{1/2} K$. Since G is positive and injective, Borel functional calculus makes G^{it} unitary, and

$$A_t = G^{it} A_0.$$

Conjugating B_0 by $\text{diag}(G^{it}, I)$ therefore gives upper-right block A_t and lower-left block A_t^* . This proves the displayed identity. $\square$

The theorem is not special to K_2 or to block-preserving couplings. In the one-sided phase ansatz, even $[G, K] \neq 0$ leaves t as a removable phase; noncommutation alone can never produce height-dependent spectrum. Moreover, a free cross-atom transition still creates temporal cycles such as $p \rightarrow q \rightarrow p$ whose lengths involve $\log p + \log q$. These are mixed primitive cycles rather than repetitions of one tensor atom. Thus a viable replacement must simultaneously use a genuinely different phase architecture that is not removable by one block unitary and supply an intrinsic homological or signed cancellation theorem preserving A1.

5.1 The exact regularized determinant and what it deletes

For the frozen rank-one binary fiber, write the off-axis formal pairing as

$$\mathcal{B}_s = \begin{pmatrix} 0 & G^s K \\ G^{1-s} K & 0 \end{pmatrix}, \quad K = \bigoplus_p K_2.$$

Each active atom block squares to p^{-1} times the rank-one projection and has nonzero eigenvalues $\pm p^{-1/2}$. The two off-diagonal pieces belong simultaneously to S_3 when

$$\frac{1}{3} < \text{Re } s < \frac{2}{3}.$$

Proposition 5.2 (Paired $\det_3$ collapse). *In the common S_3 strip,*

$$\det_3(I - z\mathcal{B}_s) = \prod_p (1 - z^2/p) e^{z^2/p}.$$

The product is independent of s . Moreover,

$$\mathrm{Tr} \mathcal{B}_s^{2r} = 2 \sum_p p^{-r}$$

converges only for $r \geq 2$, whereas odd traces vanish. The divergent quadratic trace $r = 1$ is precisely the term removed by $\det_3$.

Proof. Pair the regularized factors for the two active eigenvalues:

$$(1 - zp^{-1/2})e^{zp^{-1/2}+z^2/(2p)}(1 + zp^{-1/2})e^{-zp^{-1/2}+z^2/(2p)} = (1 - z^2/p)e^{z^2/p}.$$

Its logarithm begins at order p^{-2} , so the prime product converges. The power traces follow from the eigenvalues $\pm p^{-1/2}$. $\square$

Thus the first natural determinant reaching the critical line is both vertically sterile and amputated at the lowest nonzero trace. It receives no credit for a Riemann zero divisor. The result sharpens the next target: one needs a source-internal multi-phase chiral architecture not reducible to $A_t = G^{it} A_0$, together with exact cancellation of mixed cycles inside one trace formula. Mass noncommutation by itself does not meet this target.

6 Exact and numerical falsification audit

The CPU prototype tests algebraic identities before interpreting them. Tensor atoms up to 512 are recovered as nonunit indecomposables of the finite multiplication table; no prime list is read. The frozen run contains 97 atoms, powers $r = 1, \dots, 12$, odd Birkhoff lengths

$$31, 127, 511, 2047, 8191, 32767,$$

and a 24-point complex Mellin grid with $\mathrm{Re} s \in \{0.25, 0.5, 1, 1.5, 2, 3\}$ and $\mathrm{Im} s \in \{0, 2, 6, 12\}$. No Riemann-zero data are loaded.

Test	Frozen result	Interpretation
K_2 determinant identity	4.44×10^{-16} max residual	exact Euler block confirmed
Chiral square	1.11×10^{-16}	paired half-density confirmed
Critical-axis gauge	1.19×10^{-16}	vertical phase is removable
Chiral eigenvalues	5.55×10^{-16}	spectrum is $\{\pm p^{-1/2}\}$
Binary Mellin median error	$2.269 \rightarrow 0.04576$	improves by factor 49.6
Characteristic-function error	$5.90 \times 10^{-3} \rightarrow 5.51 \times 10^{-6}$	Gaussian channel is stable
Local-CLT max relative error	$0.0342 \rightarrow 7.62 \times 10^{-5}$	density convergence confirmed

The slow corner is reported rather than averaged away. At $s = 0.25 + 12i$, the last-cutoff Mellin relative error remains 1.56×10^4 . Negative moments emphasize the shrinking lattice points near zero, while the high imaginary part causes severe oscillation. This is consistent with pointwise and locally uniform convergence at fixed compact sets but warns that the tested cutoff is far from asymptotic in the most adversarial grid corner.

6.1 Controls that prove too much

Uniform K_3 , uniform K_4 , and a biased rank-one binary kernel all satisfy $\text{tr } K^r = 1$. A reversible binary control with second eigenvalue 0.4 instead has $\text{tr } K^r = 1 + 0.4^r$ and fails immediately. Hence the Euler ledger selects rank-one idempotence rather than the fair binary process.

After centering and variance normalization, the biased binary process and a noncanonical scalar observable on F_3 also approach the one-dimensional Gamma target. All nonconstant binary relabelings and rescalings become identical after standardization. These are honest PROVES_TOO_MUCH outcomes.

The canonical radial controls behave differently. At their final cutoffs, the K_3 radial statistic has median error 1.5793 against the binary target but 0.00996 against its native two-dimensional law. The corresponding K_4 errors are 2.5506 and 0.01009. The dimension shift predicted by [proposition 4.2](#) is therefore visible at finite cutoff.

An auxiliary Parry-inventory branch was also falsified. A pressure-normalized Hellinger identity holds for shifted and random positive inventories, hence proves too much; moreover, the identity-adjacency atom source of SD-C07 has no native nontrivial transition probability. Adding full mixing would create the mixed temporal cycles excluded by the source lock. This auxiliary branch is not part of SD-C08.

The experiment therefore supports three exact conclusions: the stationary binary fiber preserves the Euler trace ledger, the sign sector produces the Gamma skeleton, and the frozen block-preserving chiral centering is gauge-trivial. [Theorem 5.1](#) strengthens the last numerical certificate to every bounded K in the same one-sided phase ansatz. The experiment does not establish a global determinant or analytic continuation.

7 Route-A decision and Route-B lock

The preregistered positive gates P0–P5 pass: one full-shift tensor source is used; F_2 is internally selected; the Euler ledger is exact; the sign observable is a multiplicity-one representation; its Mellin limit gives the Archimedean factor; and both channels arise from the single tilted trace family of [theorem 3.2](#). We therefore assign the new candidate identifier SD-C08.

The global stop gates also fire. The Mellin and Fredholm operations are not one determinant, no internal continuation or functional equation has been proved, the chiral family has no spectral motion, and its first regularized determinant deletes the divergent quadratic trace. The stage status is

GO A3 ARCHIMEDEAN FACTOR / STOP GLOBAL COMPLETION.

7.1 Route-A layers

Layer	Verdict	Evidence
A0	analytic arithmetic origin	tensor atoms, entropy clock, and repetitions recover primes, $\log p$, and prime powers internally
A1	analytic pass	atom loops and their repetitions are exact; the binary fiber preserves rather than replaces them
A2	analytic determinant	ordinary Fredholm determinant equals $1/\zeta(s)$ on $\text{Re } s > 1$; odd Berezinian gives $\zeta(s)$
A3	partial analytic structure	source-internal Gamma factor on $\text{Re } s > 0$, but no unified determinant or continuation
A4	formal hint	centered self-adjoint chiral block exists, but it is isospectral and not a quantization

The overall decision is

ROUTE_A_ANALYTIC_CANDIDATE.

This is not ROUTE_A_SUCCESS_ROUTE_B_READY.

7.2 Why the arithmetic gate remains selective

Some local ingredients are universal: every rank-one kernel preserves unit power traces, and every suitable scalar mixing process has a Gaussian limit. SD-C08 receives A0 credit from their constrained conjunction inside the tensor source, not from either ingredient alone. The tensor monoid selects the atom loops and clock; its unique minimal atom selects F_2 ; the multiplicity-free S_2 decomposition supplies both channels; and the full q -symbol radial controls change the Gamma law. Shifted, additive, random, biased, reversible, and higher-dimensional controls reveal exactly which parts are and are not selective.

7.3 Route B

No Hilbert–Pólya operator is proposed. There is no candidate Hilbert space with a densely defined target operator, no domain or boundary conditions, no self-adjointness theorem, no compact-resolvent or counting-law result, no spectral determinant equal to a completed ξ , and no Weil-form compression. The chiral B_t is a bounded/compact diagnostic family at finite or direct-sum level, not a Route-B quantization; the universal unitary gauge collapse of the one-sided ansatz is negative evidence. Accordingly all Route-B layers are NOT_TESTABLE or false, and

route_b_invocation_allowed = false.

No coordinate from another system family is used to improve this evaluation.

8 Conclusion: the next phase architecture must change

The minimal binary full shift closes one gap left by the preceding symbolic stage. Its stationary and sign representations are two intrinsic channels of one Parry trace family. The first preserves the exact tensor-prime Euler determinant; the second generates the self-dual Gaussian and hence $\pi^{-s/2}\Gamma(s/2)$. This justifies SD-C08 and a bold but bounded A3 claim: Symbolic Dynamics now produces both the finite-prime and Archimedean factors on a common honest half-plane.

The construction also identifies its own failure mode. Every one-sided Hellinger double $A_t = G^{1/2+it}K$, including mass-noncommuting choices of K , makes $1/2$ exact but makes the vertical parameter a gauge. Regularization reaches the critical line only after deleting the first nonzero trace. Thus neither exact centering nor self-adjoint block form is enough.

The next same-family hypothesis can now be stated sharply.

Conjecture 8.1 (Cycle-cancelling multi-phase symbolic coupling). *There exists a functorially defined chiral extension of the tensor-atom fibers with at least two inequivalent t -phase channels such that:*

1. *one trace/determinant family specializes to the SD-C08 tilted Parry trace and retains every prime-power coefficient;*
2. *cross-atom mixed primitive cycles cancel by an intrinsic grading or cohomological identity, not by fitted counterterms;*
3. *its critical-axis dependence cannot be removed by a single block-diagonal unitary and is not unitarily equivalent to a constant operator;*

4. *its determinant has a nonempty analytic domain containing the symbolic Archimedean and Euler channels.*

This is a walkable next step: search within finite symbolic extensions for inequivalent phase channels, and demand exact cycle cancellation before looking at any target zeros. Merely imposing $[G, K] \neq 0$ on the present one-sided ansatz cannot work. Until the stronger theorem exists, analytic continuation, the functional equation, RH, and Route B remain outside the claim boundary.

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A Proof details, domains, and scope discipline

A.1 Local uniformity of the Mellin limit

We record the analytic point hidden by a bare appeal to the central limit theorem. Let C be a compact subset of $\{\operatorname{Re} s > 0\}$ and choose $0 < \sigma_- \leq \operatorname{Re} s \leq \sigma_+$. Split the lattice expectation into $|Y_N| < \delta$, $\delta \leq |Y_N| \leq R$, and $|Y_N| > R$.

On the middle region, the local central limit theorem gives a Riemann-sum approximation uniform in $s \in C$. Hoeffding’s binomial bound makes the tail uniformly smaller than a Gaussian moment of order $\sigma_+ - 1$ as $R \rightarrow \infty$. At the origin, the maximal lattice mass is $O(N^{-1/2})$ and the lattice spacing of Y_N is $O(N^{-1/2})$. Summing $x^{\sigma_- - 1}(1 + |\log x|^k)$ over lattice cells in $(0, \delta)$ is bounded by its integrable continuum counterpart plus a vanishing endpoint term. This controls the complex powers and their derivatives uniformly on C . First let $N \rightarrow \infty$, then $\delta \downarrow 0$ and $R \uparrow \infty$. Vitali’s theorem, or the direct derivative bounds, gives local uniformity.

The restriction to odd N is important. For even N , $S_N = 0$ has positive probability, and $|S_N|^{s-1}$ is not finite when $0 < \operatorname{Re} s < 1$.

A.2 Rank-one power-trace criterion

Suppose a finite reversible stochastic matrix K has eigenvalues $1, \lambda_2, \dots, \lambda_q$. If $\text{tr } K^r = 1$ for all $r \geq 1$, then $\sum_{j=2}^q \lambda_j^r = 0$ for all r . The first $q-1$ power sums vanish, so Newton identities make every elementary symmetric polynomial in the nonstationary eigenvalues zero. Hence all $\lambda_j = 0$. Reversibility makes K diagonalizable in its stationary inner product, so it is a rank-one projection. This proves that the Euler ledger tests rank one, not alphabet size.

A.3 Operator ideals for the chiral pair

The singular values of the upper-right atom blocks scale as $p^{-\text{Re } s}$, and those of the lower-left blocks as $p^{-(1-\text{Re } s)}$. Both lie in S_q exactly when

$$q \text{Re } s > 1, \quad q(1 - \text{Re } s) > 1.$$

For $q = 2$ this gives the open strip $1/2 < \text{Re } s < 1/2$, which is empty. The first integer class with a nonempty common strip is $q = 3$, giving $1/3 < \text{Re } s < 2/3$. The $\det_3$ subtraction removes trace powers one and two. Odd chiral traces already vanish, while the power-two trace is the divergent $2 \sum_p p^{-1}$. The resulting determinant therefore cannot be credited with preserving the low-order Euler ledger.

A.4 Data and interpretation firewall

The candidate definition uses no Riemann-zero list, no fitted critical-axis scale, and no target-root loss. Prime atoms are recovered as tensor indecomposables, not read from a prime file. The $1/2$ appearing in the Gaussian channel is the diffusive exponent and the $1/2$ appearing in the chiral channel is the equality point of paired exponents s and $1-s$. Neither produces the Riemann functional equation by itself.

The following implications are forbidden:

$$\begin{aligned} \text{Gamma factor} &\not\Rightarrow \text{analytic continuation,} \\ \text{self-adjoint blocks} &\not\Rightarrow \text{Hilbert-Pólya,} \\ \text{critical centering} &\not\Rightarrow \text{vertical zeros.} \end{aligned}$$

Any idea requiring a different primary system family is excluded from the paper and may only be stored as a future ROUND2_CLUE.